

Rithmomachia – Quick Reference

White has the first turn, after that they turns alternate between players. Each turn has 3 parts: (1) a pre-move capture, (2) a piece move, (3) a post-move capture.

For parts (1) and (3): if you don't have any captures or have some but don't want to make them yet, that's okay you can skip it.

For (2): you must make a move. If you have no legal moves you lose.

Setup, Pieces & Notation

Notes - Movement path/landing must be legal for the capturing piece. No jumping or diagonal moves or landing on another piece.

Pieces & typical roles

- ○ **Circle** — moves 1 orthogonally (up-down or left-right)
- △ **Triangle** — moves 2 orthogonally (no jumping)
- □ **Square** — moves 3 orthogonally (no jumping)
- **Pyramid (stack)** — moves 3 orthogonally (no jumping). The value equal to the sum of stacked square subpiece values.

Coordinates & quick values

- **Initial Position:**

	1	2	3	4	5	6	7	8
p	49	121	-	-	-	-	225	361
o	28	66	36	30	56	64	120	190
n	16	12	9	25	49	81	90	100
m	-	-	3	5	7	9	-	-

...middle rows empty at start...

d	-	-	8	6	4	2	-	-
c	81	72	64	36	16	4	6	9
b	153	91	49	42	20	25	45	15
a	289	169	-	-	-	-	81	25

Captures & Victories

Arithmetic capture patterns (use only **your** piece values on the LHS; result must equal the **enemy** piece's value)

- **Number:** ($A = C$) and friendly piece of value A is *one* move from the enemy piece of value C.
- **Sum:** ($A + B = C$) and friendly pieces of values A, B are each *one* move from the enemy piece of value C.
- **Difference:** ($B - A = C$, with $A < B$) and friendly pieces of values A, B are each *one* move from the enemy piece of value C.
- **Product:** ($A \cdot d = C$, integer $d > 0$) and friendly piece of value A is d coordinates from the enemy piece of value C (with no pieces in between them).
- **Divisor:** ($A / d = C$, integer $d > 0$) and friendly piece of value A is d coordinates from the enemy piece of value C (with no pieces in between them).
- **Siege** (positional): Enclose or surround an enemy's position.

Progression tests (for patterns & wins)

- **Arithmetic progression (AP):** ($a, a+d, a+2d$) with common difference (d). Example: **3, 5, 7**.
- **Geometric progression (GP):** (a, ar, ar^2) with ratio ($r > 1$). Example: **2, 4, 8**.
- **Harmonic progression (HP):** ($1/a, 1/b, 1/c$) form an AP. Example: **9, 15, 45**.

Victory conditions (name & prove)

- **Proper victory:** Exhibit a valid **AP / GP / HP** with your pieces that are all in enemy territory.
- **Common victory:** Achieve a capture condition first (e.g., 1) first player to capture 10 enemy pieces wins, or 2) first player to capture enemy pieces totally over 400 in value wins).

Pyramid stacks (initial setup)

- **White:** pyramid subpieces **1, 4, 9, 16, 25, 36** → base square **91**.
- **Black:** pyramid subpieces **16, 25, 36, 49, 64** → base square **190**.
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values	circles	triangles	squares
white	2, 4, 4, 4, 6, 8, 16, 36, 64	6, 9, 20, 25, 42, 49, 72, 81	15, 25, 45, 81, 91 , 153, 169, 289
black	3, 5, 7, 9, 9, 25, 49, 81	12, 16, 30, 36, 56, 64, 90, 100	28, 66, 120, 121, 190 , 225, 361